EXPERIMENT-01

AIM

To write a Python program using NumPy to calculate the average score of each subject from a student score matrix and identify the subject with the highest average score.

ALGORITHM

1. Start the program.
2. Import the NumPy library.
3. Create a 4×4 NumPy array where each row represents a student and each column represents a subject.
4. Store subject names in a list.
5. Compute the average score of each subject using np.mean() with axis=0.
6. Identify the index of the highest average score using np.argmax().
7. Find the subject corresponding to the highest average.
8. Display the average scores and the subject with the highest average.
9. End the program.

PROGRAM

```
import numpy as np

# 4x4 matrix → each row = student, each column = subject

# Columns: Math, Science, English, History

student_scores = np.array([

    [85, 78, 92, 74],

    [88, 90, 85, 80],

    [76, 85, 88, 90],

    [90, 92, 84, 86]

])

# Calculate average for each subject (column-wise mean)

subject_averages = np.mean(student_scores, axis=0)
```

```
subjects = ["Math", "Science", "English", "History"]

print("Average Score for Each Subject:")
for sub, avg in zip(subjects, subject_averages):
    print(f"{sub}: {avg:.2f}")

# Find highest average
highest_avg_index = np.argmax(subject_averages)
print("\nSubject with Highest Average Score:", subjects[highest_avg_index])
```

OUTPUT:

```
Average Score for Each Subject:
Math: 84.75
Science: 86.25
English: 87.25
History: 82.50
Subject with Highest Average Score: English
```

RESULT

The program successfully calculated the average score for each subject and identified the subject with the highest average score using NumPy operations.